

SpaceX Falcon 9 First Stage Landing Prediction

IBM Data Science Professional Certificate
Capstone Project

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OUTLINE



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EXECUTIVE SUMMARY



- **Objective:** Predict Falcon 9 first stage landing success using historical launch data.
- **Data Sources:** SpaceX REST API and Wikipedia.
- **Data Preparation:** Cleaned, merged and transformed the dataset.
- **Analysis:** EDA, SQL, Folium maps and Plotly Dash dashboard.
- **Predictive Modeling:** Built classification models and evaluated their performance.
- **Key Outcome:** Identified factors affecting landing success and achieved accurate predictions.



INTRODUCTION



- **Project Background:** SpaceX reduces launch costs by reusing Falcon 9 first-stage boosters.
- **Business Problem:** Predict whether the Falcon 9 first-stage landing will be successful before launch.
- **Project Objective:** Build and evaluate a machine learning classification model using historical SpaceX launch data.
- **Dataset:**
 - SpaceX REST API
 - Wikipedia launch records

METHODOLOGY



- **Data Collection:**
 - Collected launch data from **SpaceX REST API**
 - Retrieved launch records from **Wikipedia**
- **Data Wrangling:**
 - Cleaned missing values
 - Merged datasets and selected relevant features
- **Exploratory Data Analysis (EDA):**
 - Performed data visualization using **Matplotlib** and **Seaborn**
 - Analyzed factors affecting landing success
- **Interactive Analysis:**
 - Executed **SQL queries** for data exploration
 - Created **Folium maps** and **Plotly Dash dashboard**
- **Predictive Modeling:**
 - Built classification models
 - Evaluated models and selected the best performer

RESULTS

EDA and machine learning analyses were performed on historical Falcon 9 launch data.

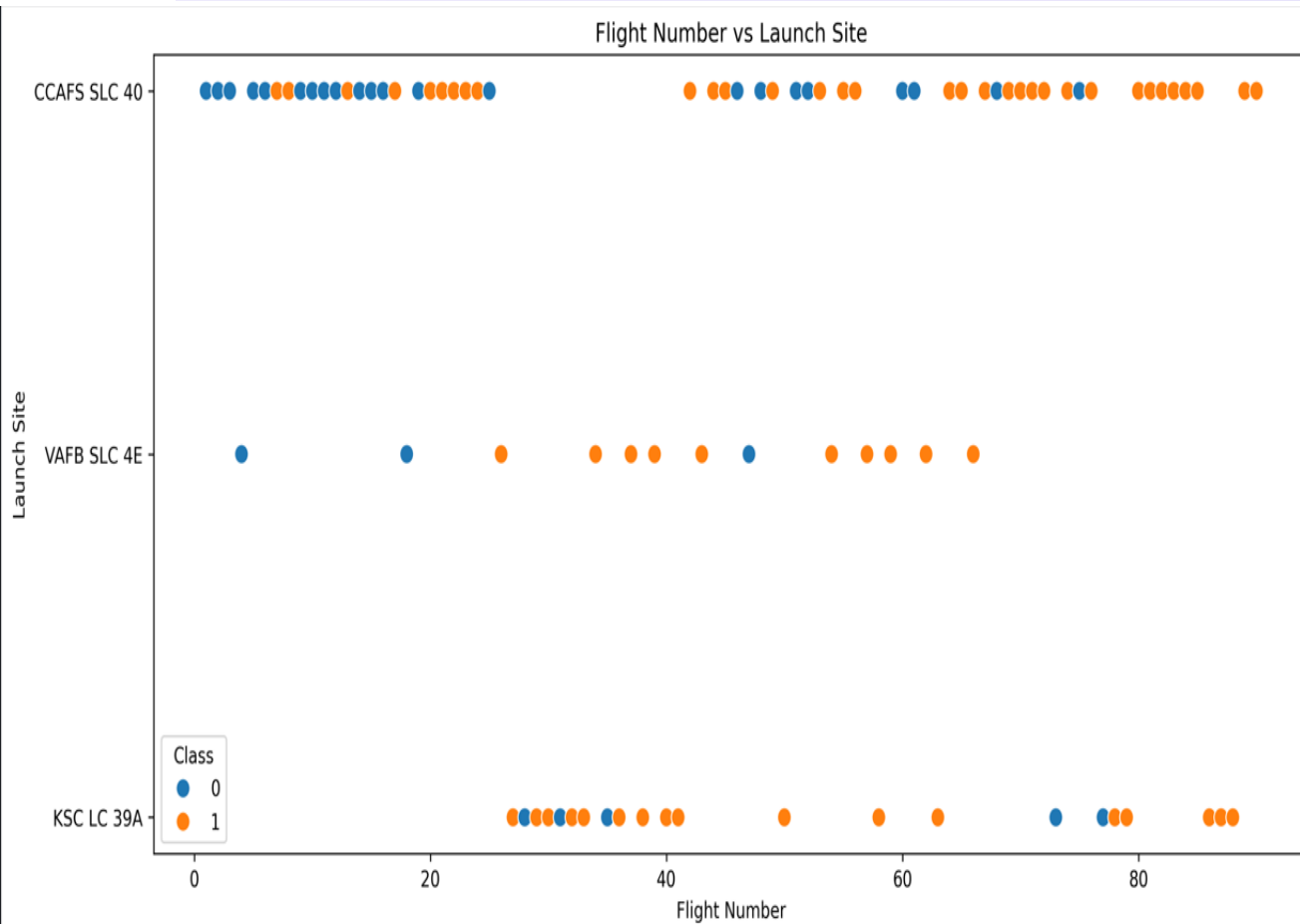
Landing success was influenced by **launch site, payload mass, orbit type, and flight number.**

Launch site proximity and **yearly landing success trends** revealed important mission patterns.

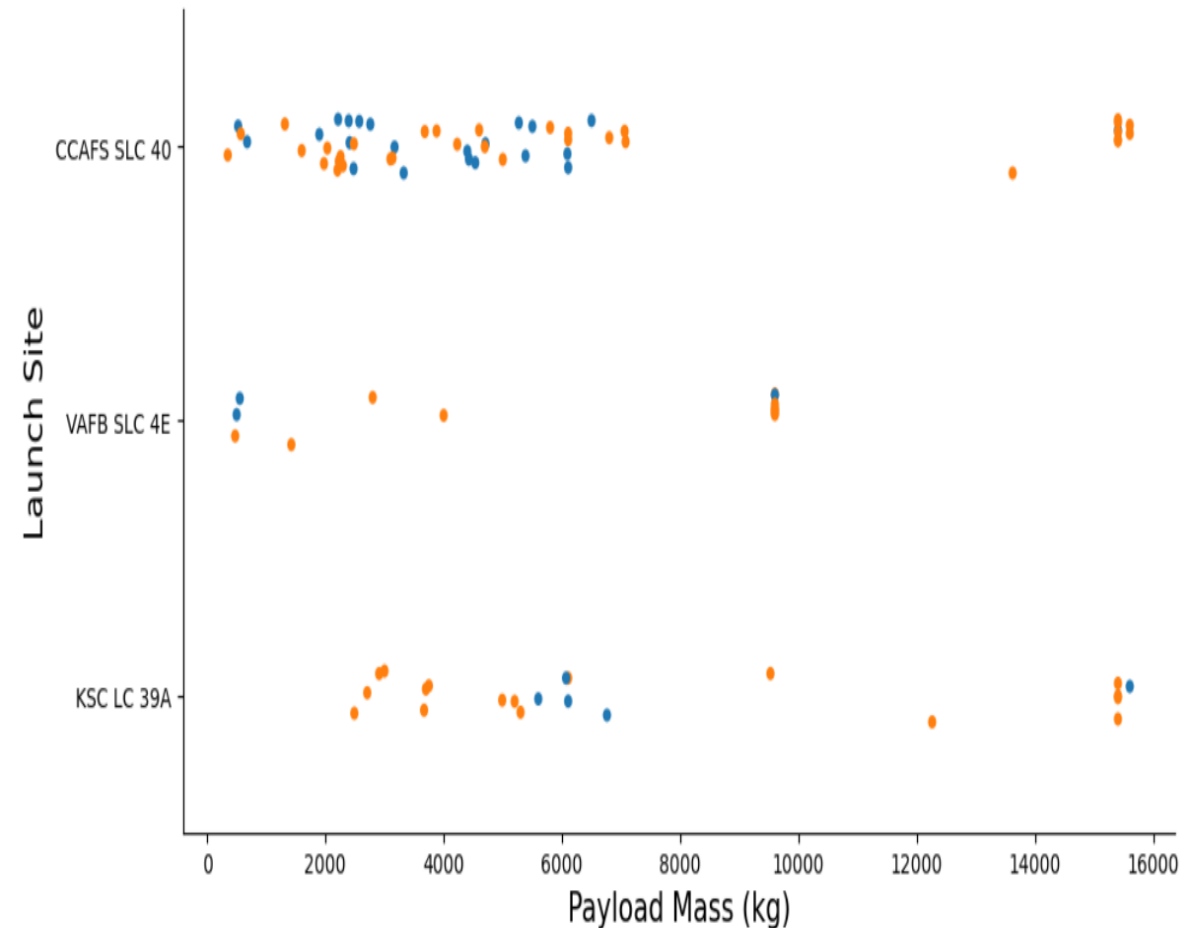
The findings support the development of accurate predictive models for Falcon 9 landing success.



Flight Number vs. Launch Site



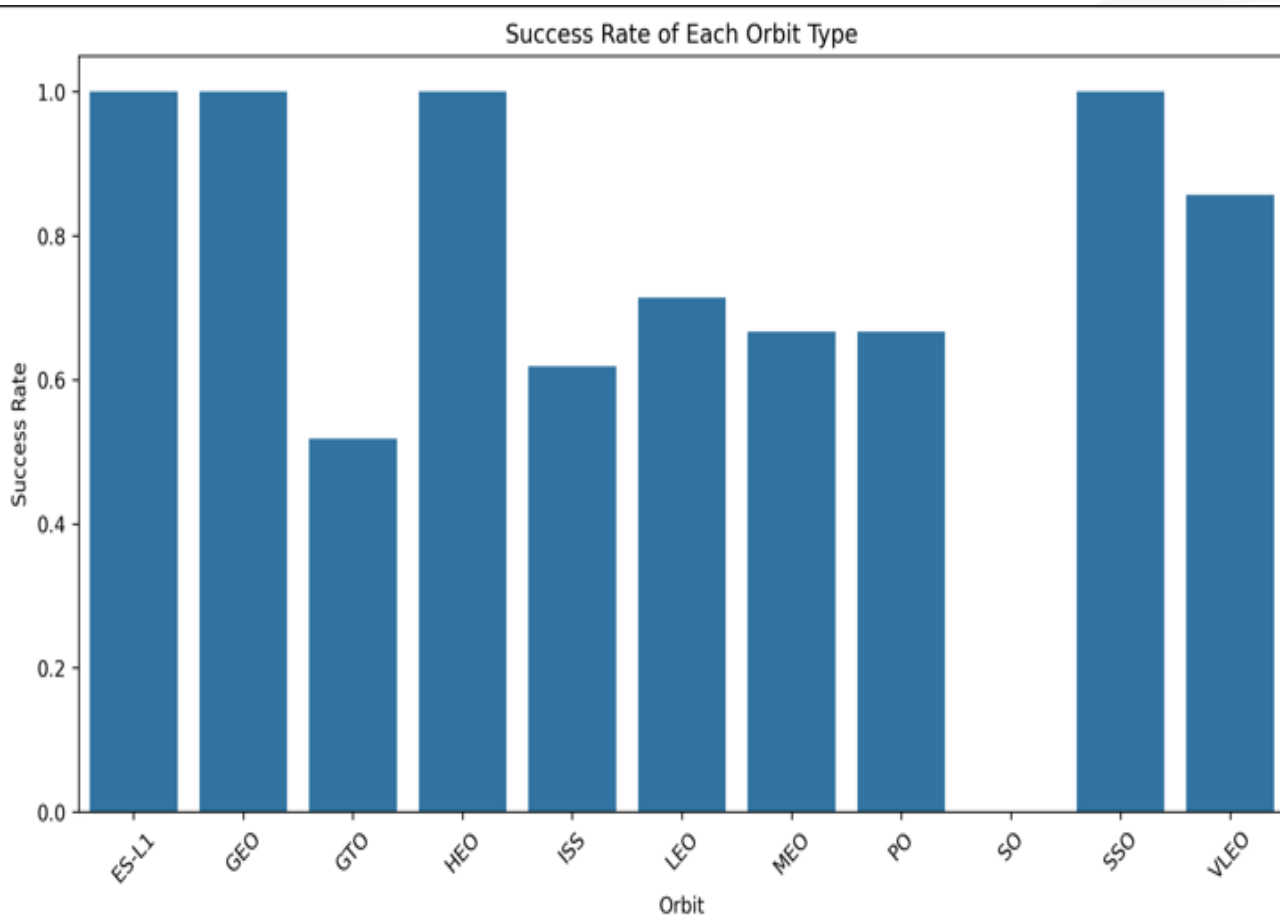
Payload Mass vs. Launch Site



• Key Findings

- CCAFS SLC 40 handled launches across a wide range of payload masses.
- KSC LC 39A was mainly used for medium to high payload missions.
- VAFB SLC 4E had fewer launches than the other launch sites.
- Successful and failed landings occurred across all payload mass ranges, indicating that payload mass alone does not determine landing success.

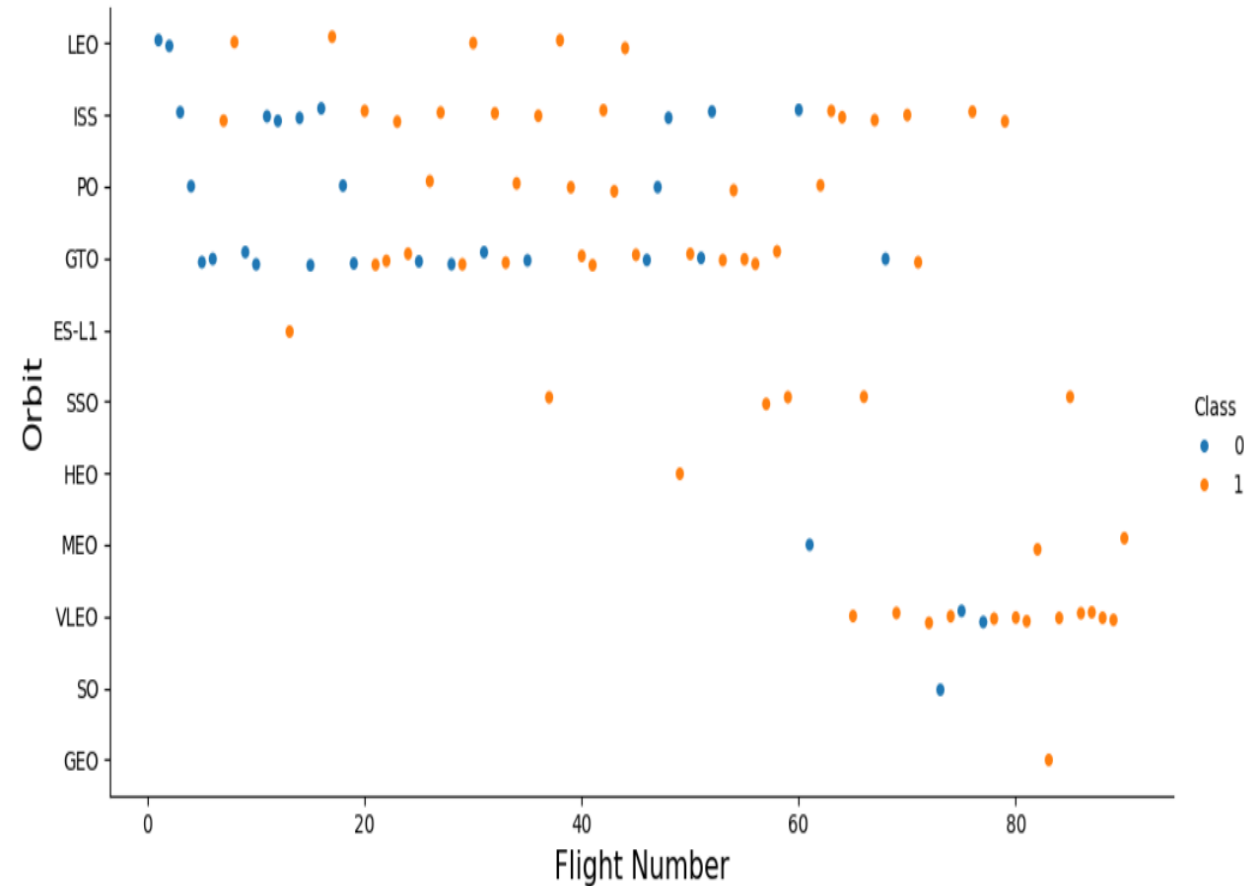
Success Rate by Orbit



- **Key Findings**

- ES-L1, GEO, HEO and SSO achieved 100% success.
- VLEO achieved a high success rate (~86%).
- GTO had a comparatively lower success rate.
- SO recorded no successful landings in the available dataset.

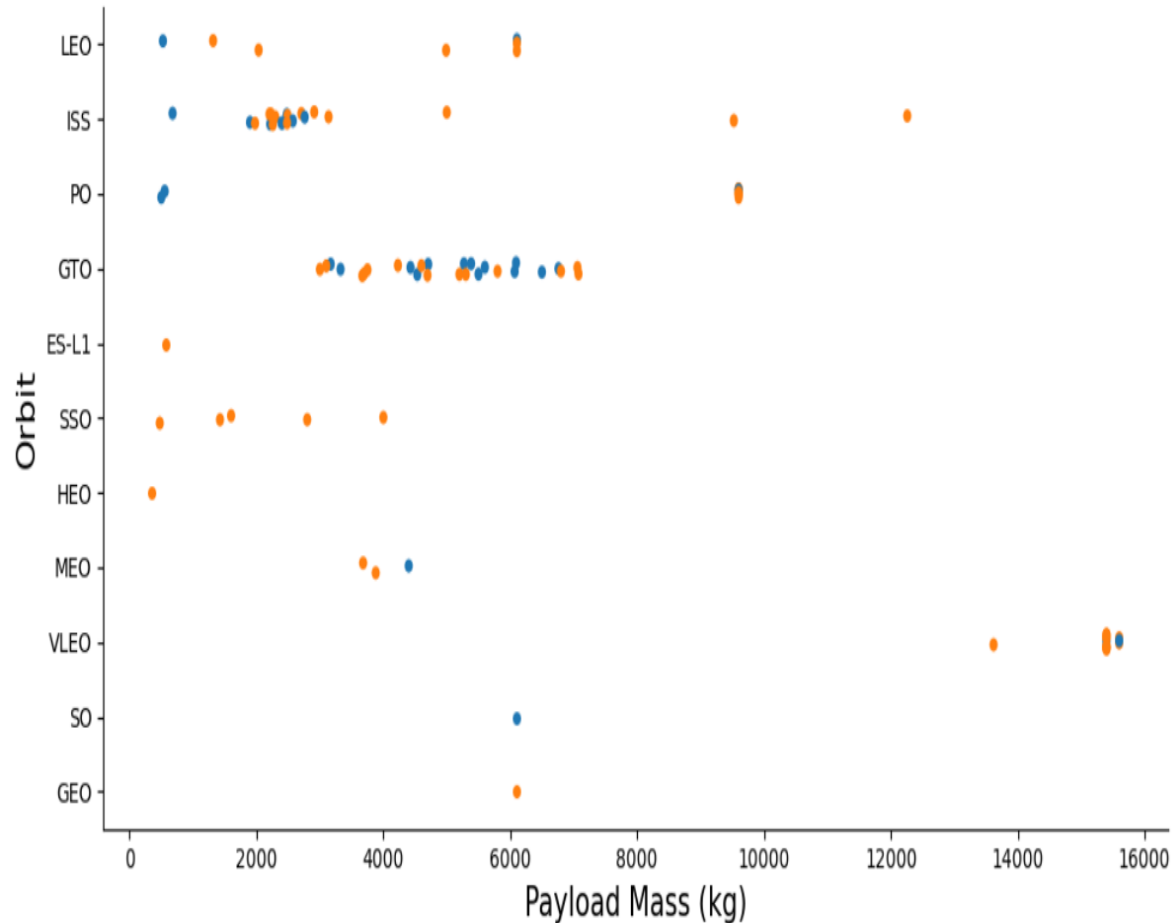
Flight Number vs. Orbit



• Key Findings

- LEO, ISS, and VLEO missions generally showed high landing success across many flights.
- GTO missions included both successful and failed landings.
- Later flight numbers generally showed improved landing success across multiple orbit types.
- Orbit type appears to influence the likelihood of a successful landing.

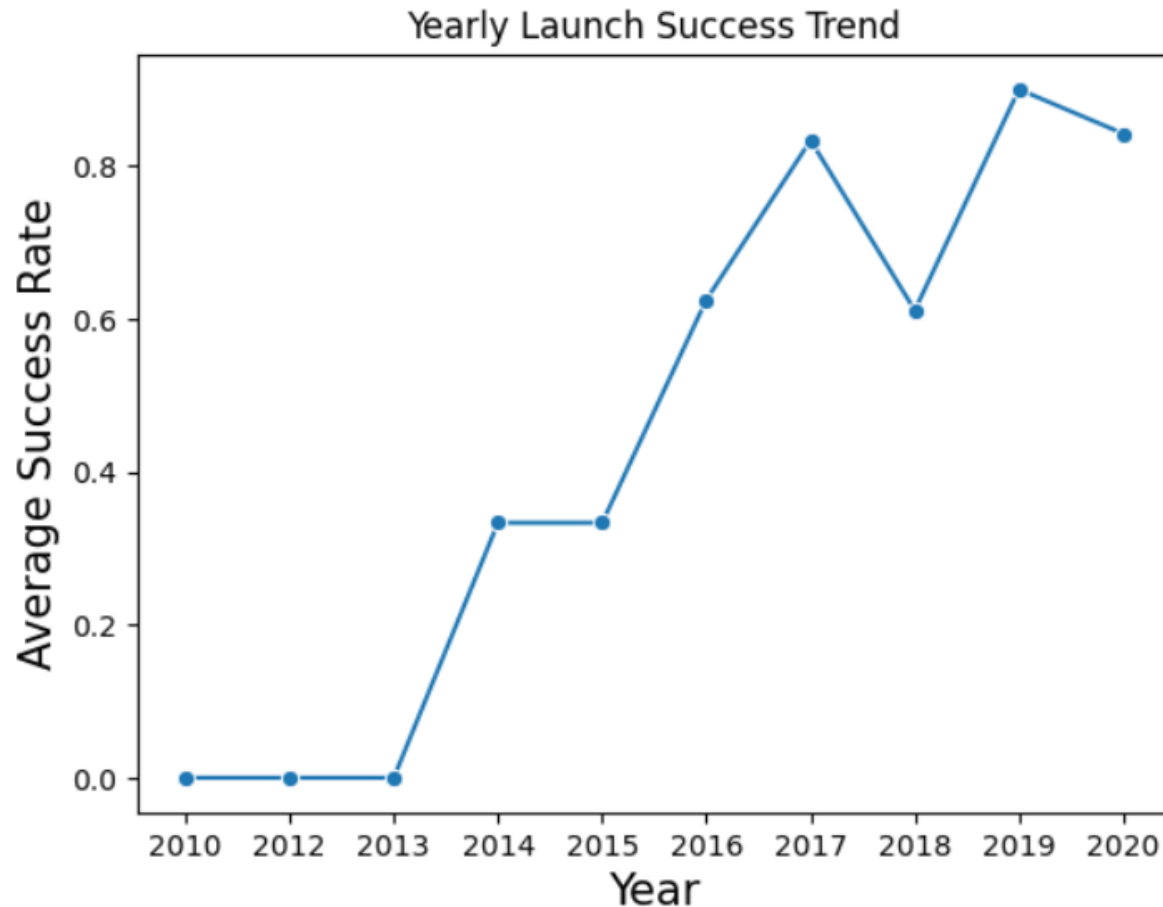
Payload Mass vs. Orbit



• Key Findings

- GTO missions were conducted across a broad range of payload masses.
- VLEO missions were associated with the highest payload masses.
- LEO and ISS missions included low to medium payload missions.
- Payload mass varied across orbit types, suggesting that orbit type is an important factor in landing outcome analysis.

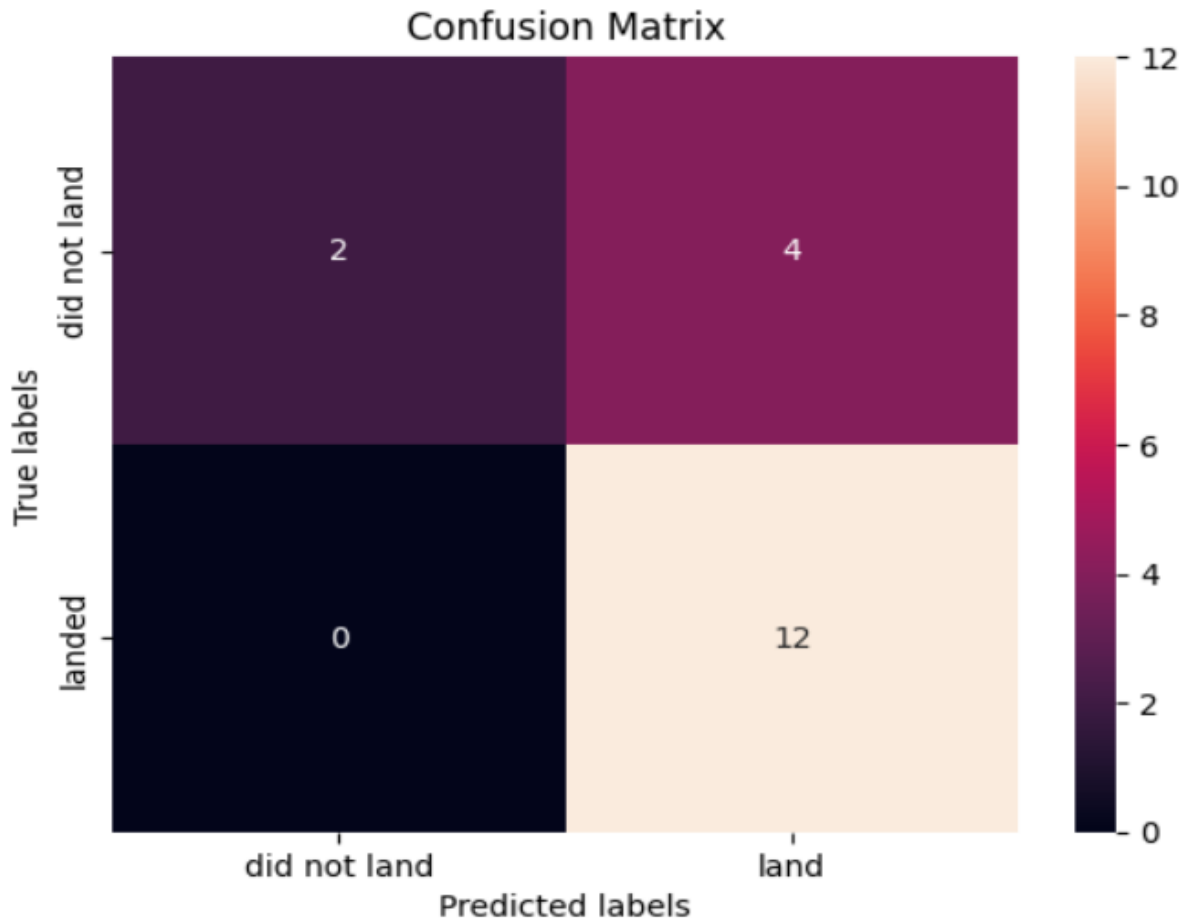
Yearly Launch Success Trend



• Key Findings

- Overall launch success rates improved over the years despite minor fluctuations.
- Success rates were very low during the early years (2010–2013).
- From 2016 onwards, the average success rate increased significantly.
- By 2019–2020, SpaceX achieved consistently high launch success rates.

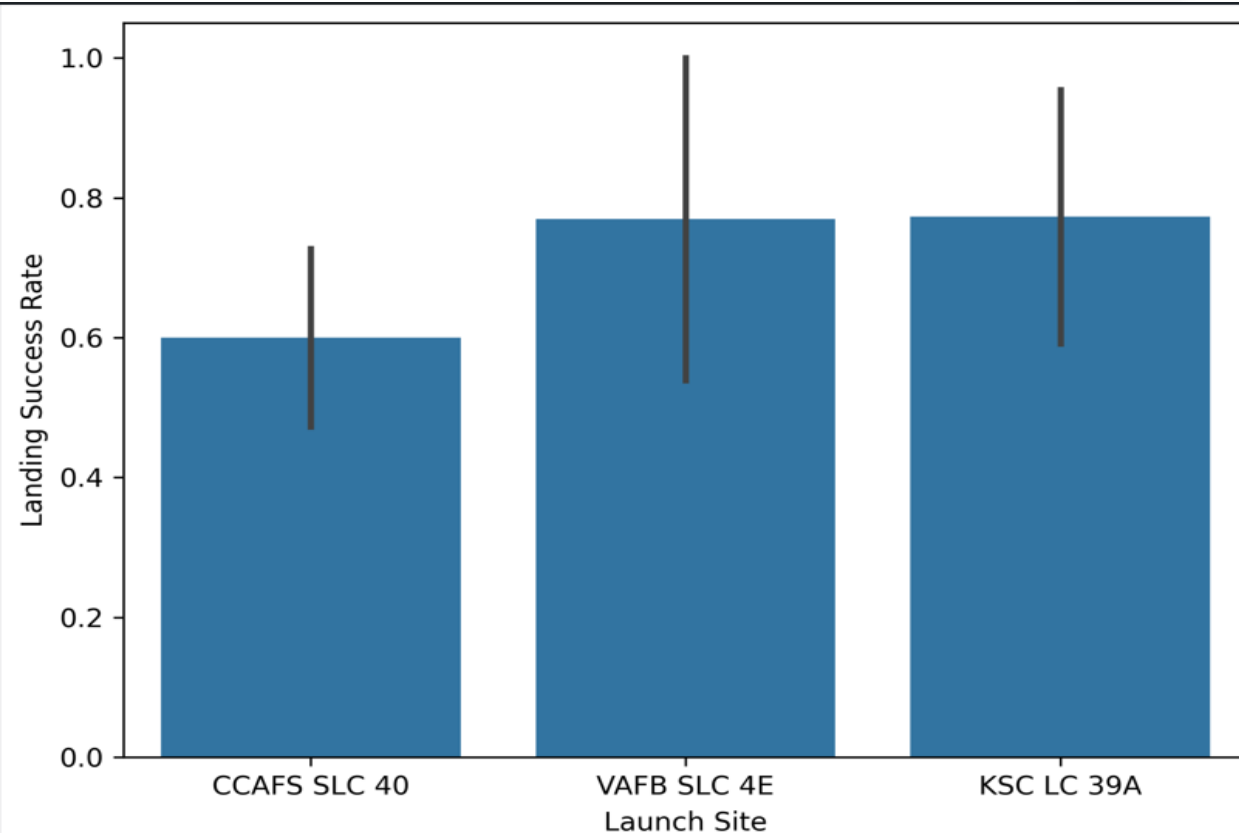
Model Evaluation – Confusion Matrix



- **Key Findings**

- The model correctly predicted **12 successful landings (True Positives)**.
- **2 unsuccessful landings** were correctly classified (True Negatives).
- **4 unsuccessful landings** were incorrectly predicted as successful (False Positives).
- The model demonstrates strong performance in predicting successful landings, with a small number of false positive predictions.

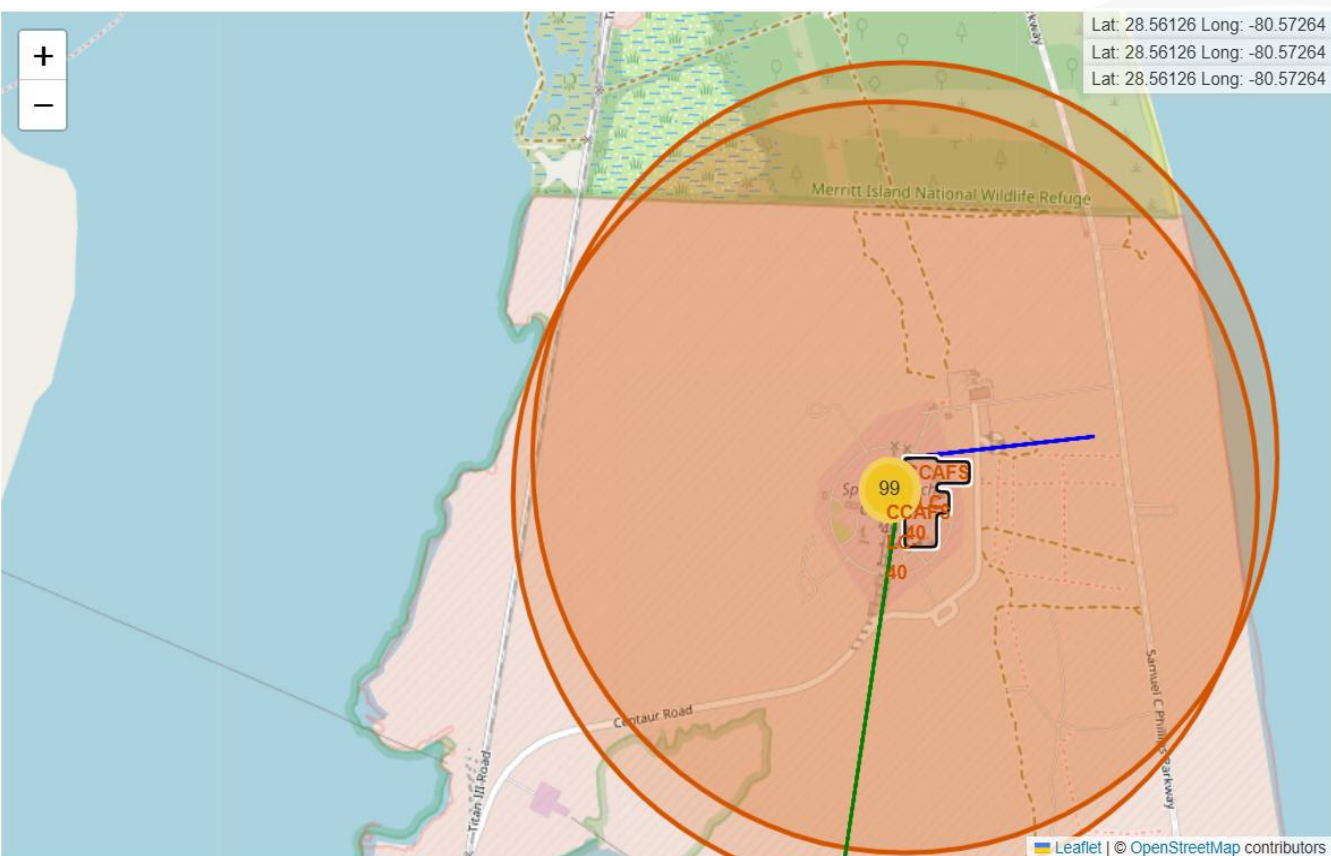
Landing Success by Launch Site



• Key Findings

- KSC LC 39A recorded the highest landing success rate.
- VAFB SLC 4E also demonstrated a strong success rate.
- CCAFS SLC 40 showed a comparatively lower average success rate.
- Launch site appears to influence landing success.

Launch Site Proximity Analysis (CCAFS)



• Key Findings:

- CCAFS launch site is located close to the coastline.
- The coastal location supports safe launch trajectories over the ocean.
- The site is positioned away from major populated areas to improve safety.
- The location provides a safe launch corridor over the Atlantic Ocean.

SPACEX FALCON 9 – FINDINGS & IMPLICATIONS

Findings

- Launch success rate improved over time.
- Launch site and payload significantly affected landing success.
- The selected classification model demonstrated good predictive performance.

Implications

- Machine learning can effectively predict Falcon 9 landing success.
- The prediction model can support mission planning and cost reduction.
- Data-driven insights help improve future SpaceX launch decisions.



RESULTS – VISUALIZATIONS & DASHBOARD

EDA Charts

- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Success Rate by Orbit
- Flight Number vs. Orbit
- Payload Mass vs. Orbit
- Yearly Launch Success Trend
- Landing Success by Launch Site

Interactive Dashboard

- SQL Query Results
- Folium Launch Site Map
- Plotly Dash Dashboard
- Machine Learning Prediction Dashboard

Conclusion

• Key Takeaways

- The analysis identified several factors that influence **SpaceX Falcon 9 landing success**.
- **Flight Number** showed a positive relationship with landing success, indicating improved performance over time.
- **Payload Mass, Orbit Type, Launch Site, and Booster Version** all had a measurable impact on mission outcomes.

• Overall Findings

- The **yearly trend** demonstrated a significant improvement in landing success rates, with By 2019, launch success rates reached their highest levels and remained consistently high thereafter.
- Overall, the analysis indicates that SpaceX has significantly improved Falcon 9 landing reliability through technological advancements and operational experience.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Launch success rate improved significantly over the years.
- Flight number, launch site, orbit type, and payload mass were identified as key factors influencing landing success.
- Machine learning models successfully predicted Falcon 9 landing outcomes with good accuracy.

Implications

- Predictive models can support mission planning and decision-making.
- Understanding launch factors helps improve mission reliability and reduce operational costs.
- Data-driven analysis enables better forecasting of future Falcon 9 landing success.

DASHBOARD



<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>



DISCUSSION



- **Key Discussion Points**

- Launch success was influenced by payload mass, launch site, and orbit type.
- EDA identified important trends in the SpaceX launch dataset.
- Plotly and Folium visualizations improved interactive data exploration.
- The classification model achieved good predictive performance.
- Historical data may limit prediction accuracy for future missions.
- Future work can include weather data, real-time launch conditions, and more advanced machine learning models.

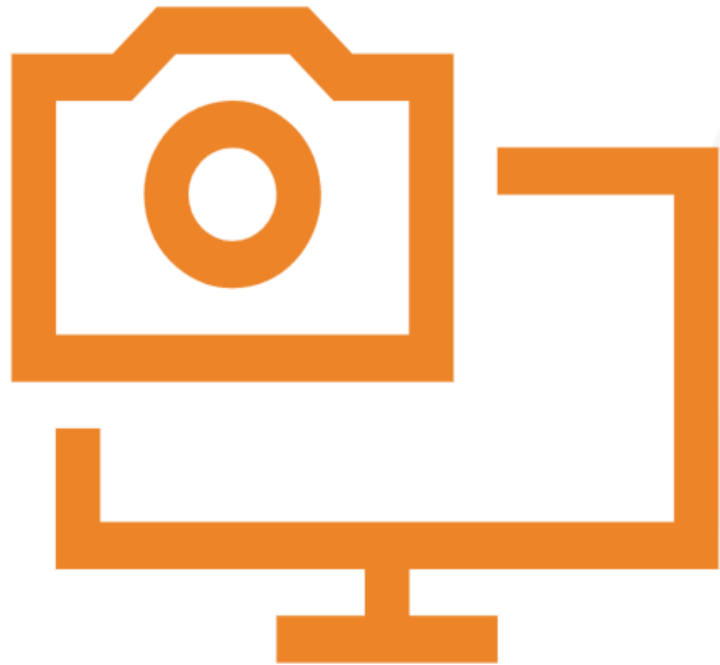


CONCLUSION



- Launch success increased over time.
- Payload and orbit influence mission success.
- The classification model achieved good predictive performance.
- The interactive dashboard supports data exploration.

APPENDIX



- **Additional Analysis**
- SQL query outputs used for exploratory analysis.
- Interactive Plotly dashboard for launch prediction.
- Folium map showing launch site proximity.
- Additional EDA charts supporting model development.